

Whether protein consumption can influence human height?

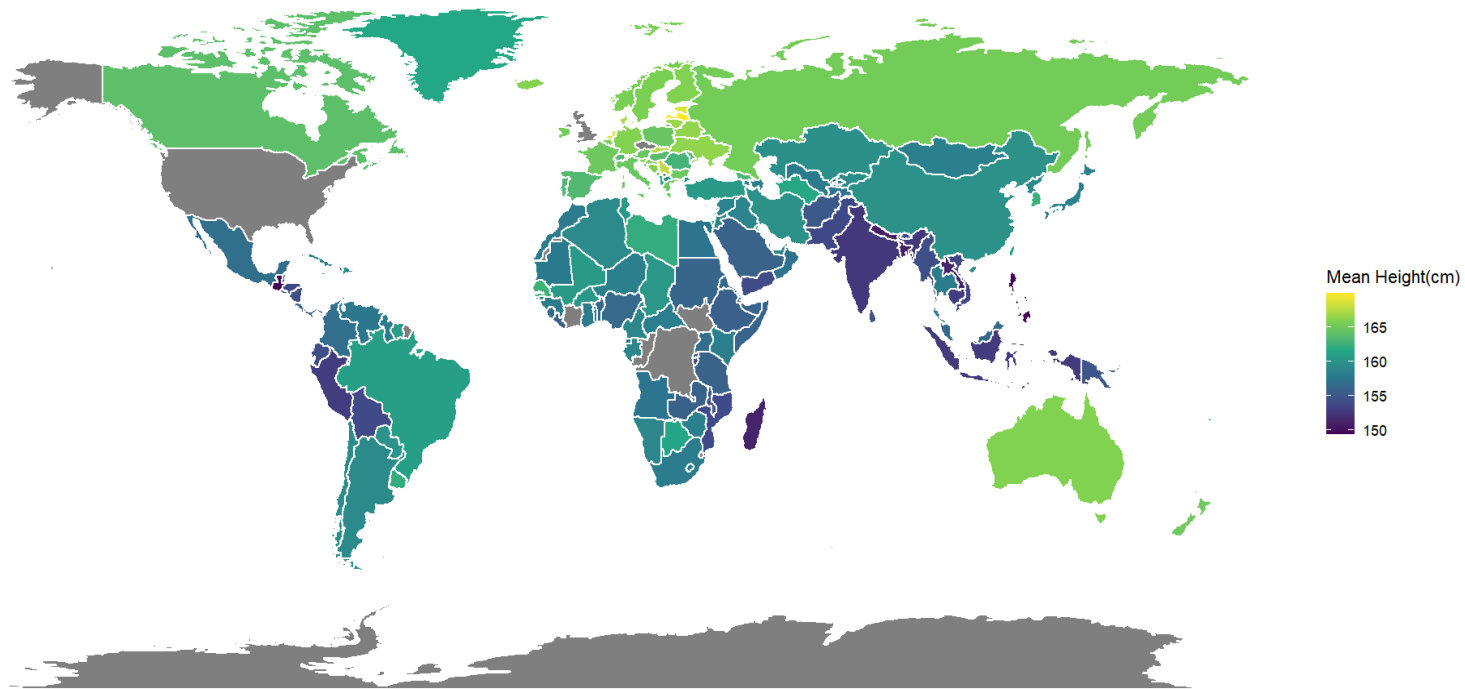
Course Code: [REDACTED]

Word Count: 3,192

Registration Number: [REDACTED]

Graph 1-1

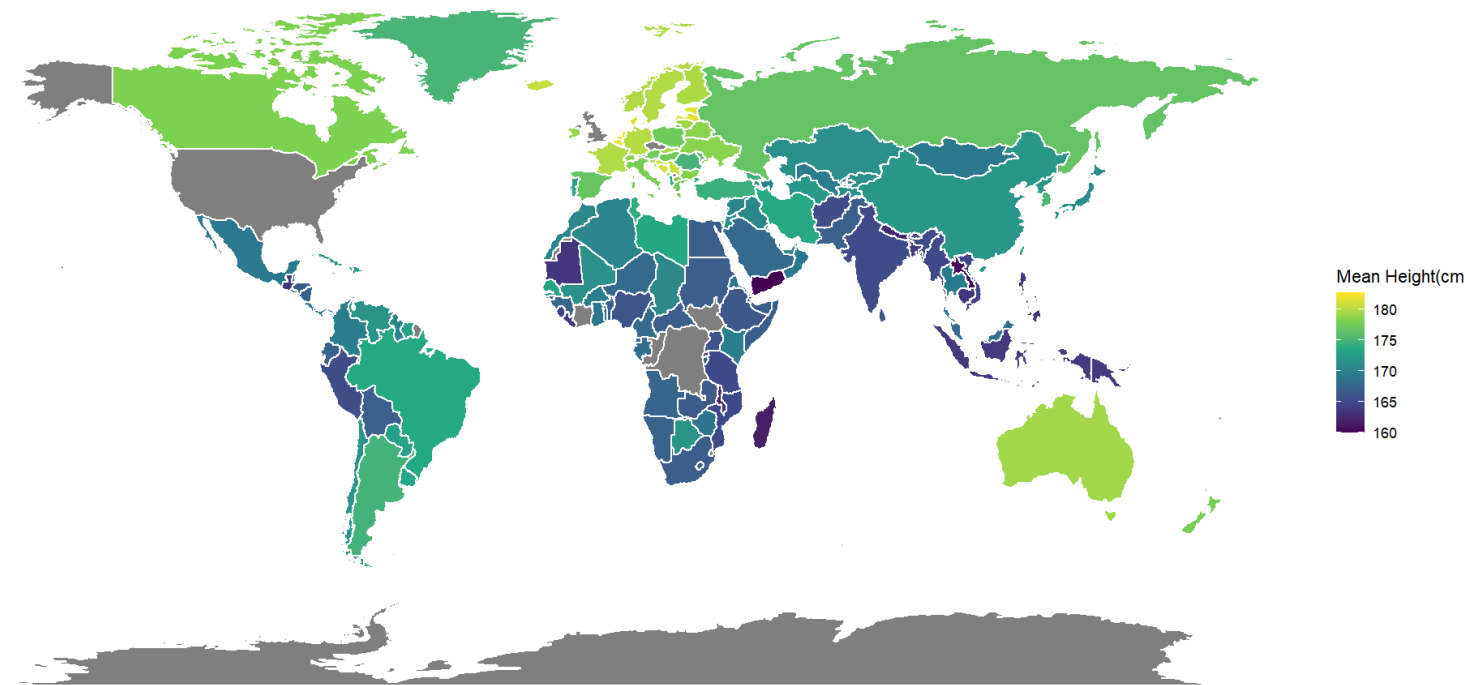
Mean height of women aged 18 in 2014



Graph 1-2

Mean height of men aged 18 in 2014

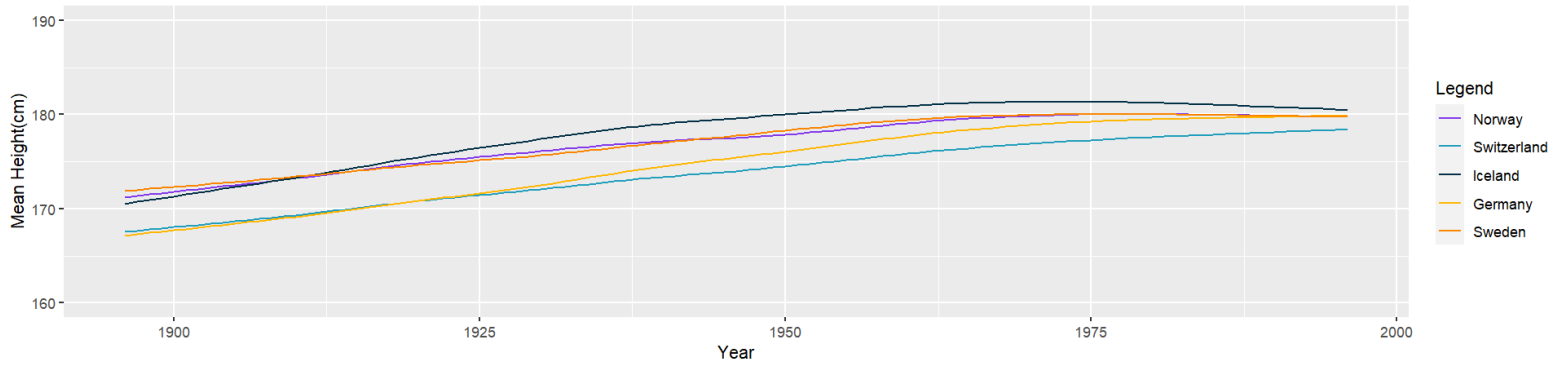
NCD RiskC, Human Height (2017)



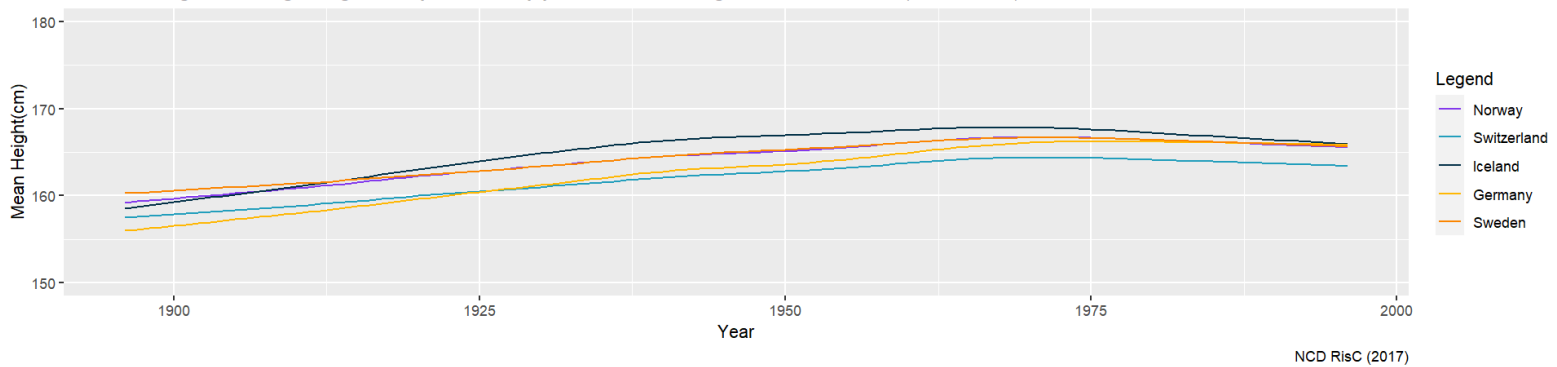
NCD RiskC, Human Height (2017)

Graph 2

Annual change in average height of 18y males by year of birth in 5 highest HDI countries (1896~1996)

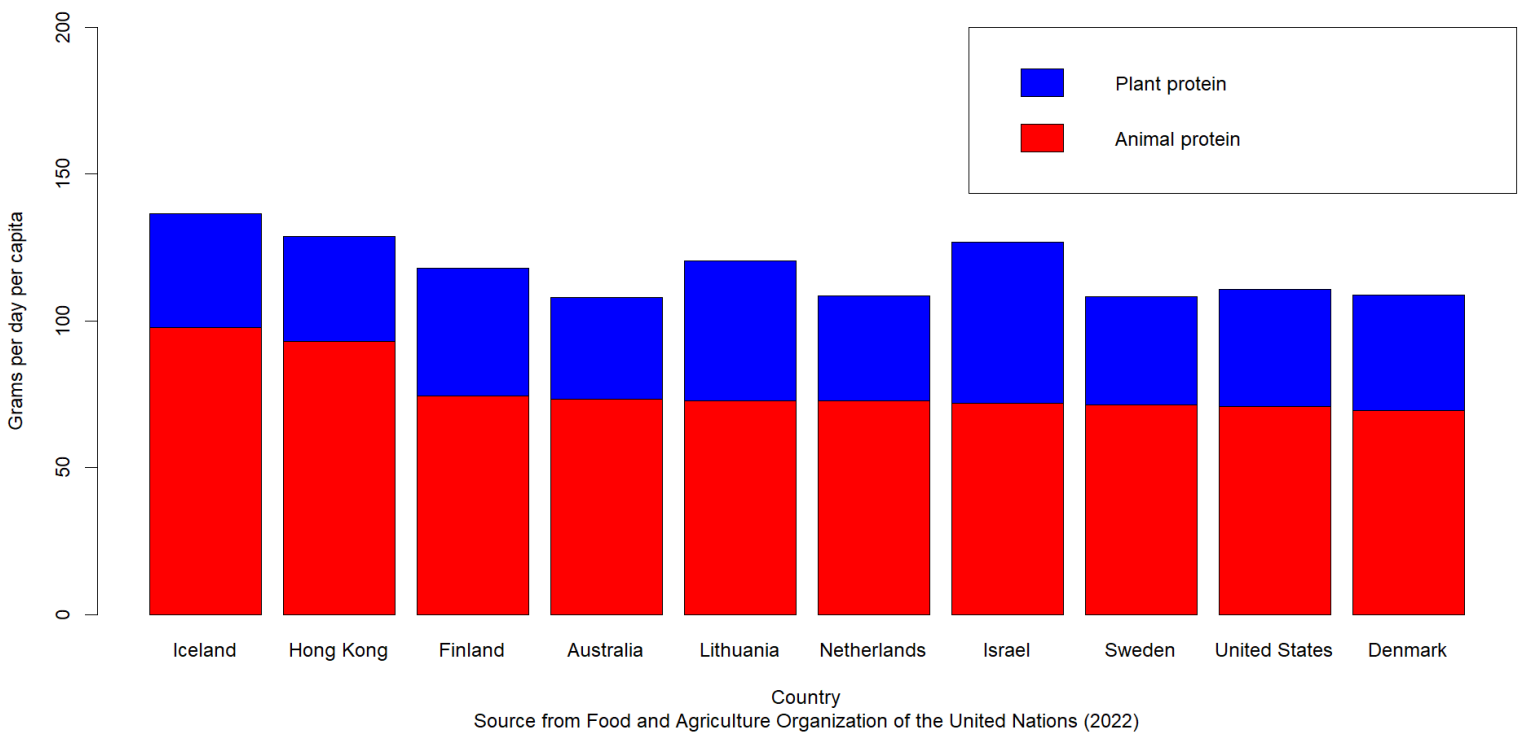


Annual change in average height of 18y females by year of birth in 5 highest HDI countries (1896~1996)



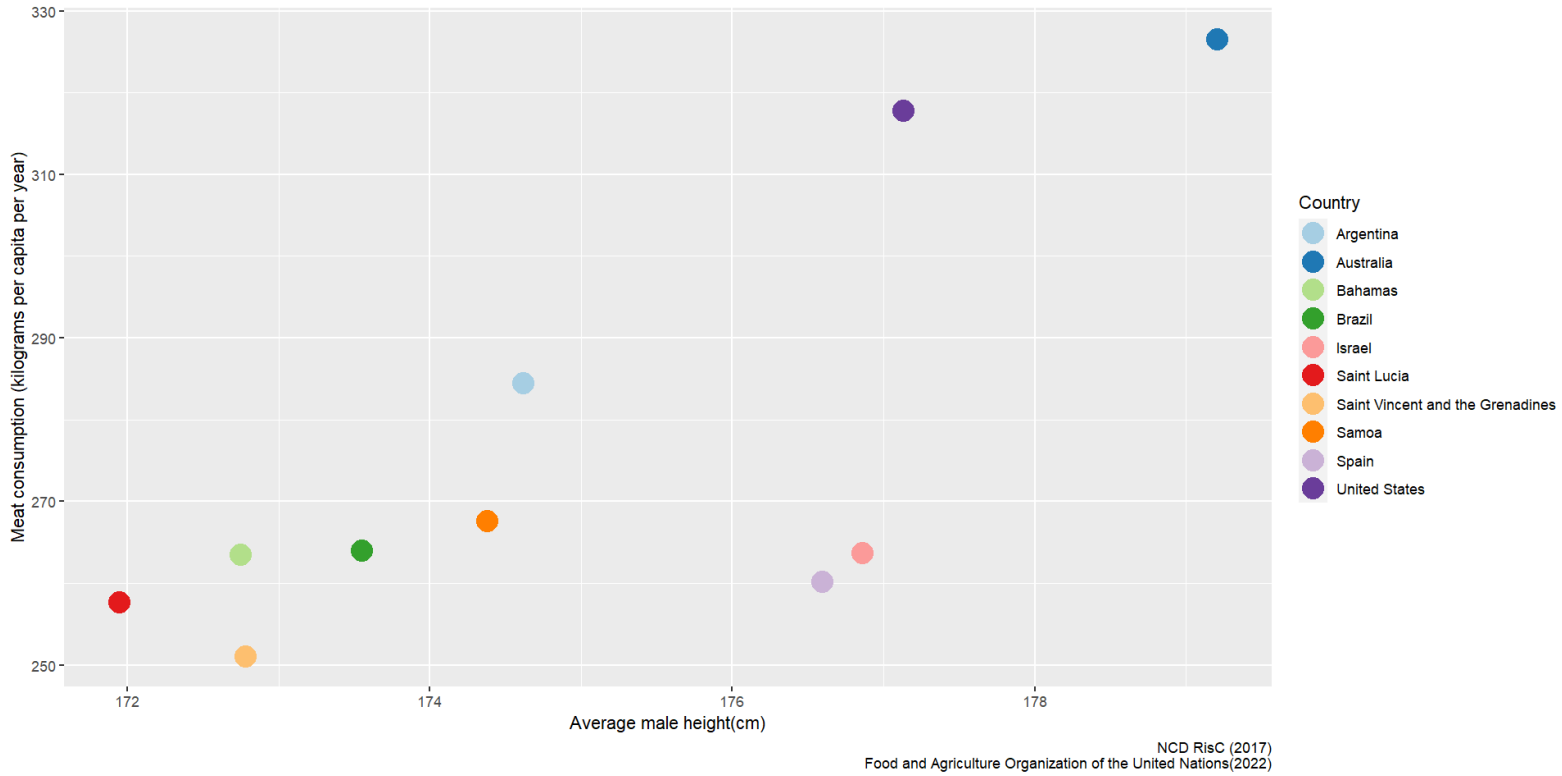
Graph 3

Daily protein supply from animal and plant-based foods in 2014



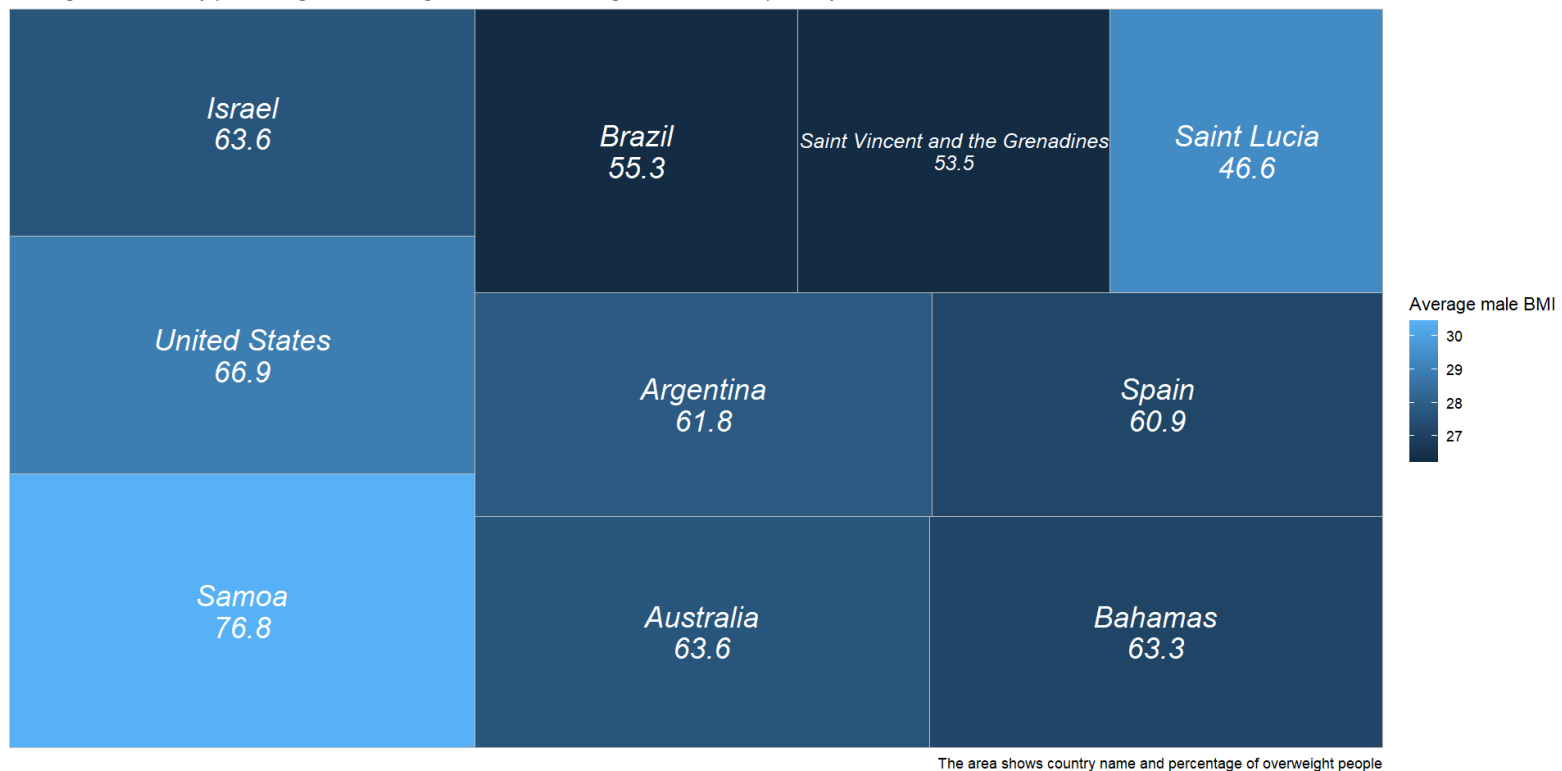
Graph 4

Average male height by top 10 countries where men ate the most meat in 2014



Graph 5

Average male BMI by percentage of overweight in countries of high meat consumption by males in 2014



Knowledge building

Human development has always been an important issue, and it is valuable for researchers to predict the future generation's growth. In this report, I want to find out whether the human height trend and the protein consumption are related or not.

When we were young, we always listened to our parents to drink milk in order to grow taller. Toddlers who eat eggs and milk are taller than those who do not (Mosites et al., 2017). However, there is an opposite statement saying that there is no conclusive evidence showing that dairy product intake and height are related whereas children without having milk are more likely to have lower bone mineral densities and more fractures (de Lamas et al., 2019).

What if some people cannot drink milk? For instance, some people may be allergic to milk, and some people are lactose intolerance. Moreover, the increasing vegan, vegetarian, and flexitarian populations are desired to plant proteins (Ismail et al., 2020). Animal proteins involve milk, cheese, meat, fish and eggs while plant proteins are mainly from 'potatoes and other tubers, vegetables, legumes, fruits, nuts and seeds, cereal and cereal products, and soya products' (Lin et al., 2011). There is further evidence that children who drink non cow milk beverages such as soy and almond milk are shorter than others who drink milk (Morency et al., 2017).

Animal proteins can be digested easier than plant proteins (Berrazaga et al., 2019). However, consuming animal protein may lead to high risk of obesity (Segovia-Siapco, 2019 & Bujnowski, 2011), this is because of red and processed meat (Halkjær et al., 2011).

According to the above literature review, I would like to examine whether the recent trend of human height and the consumption of protein have a relationship by showing different kinds of graphs. I use some datasets from Our World in Data (<https://ourworldindata.org>). There is a topic talking about human height such as average height in men and women (Rodriguez-Martinez et al., 2020). Nevertheless, I not only use the dataset from this topic, but also use the topics such as Obesity (Abarca-Gómez et al., 2017), Food Supply, and Meat and Dairy Production (Food and Agriculture Organization of the United Nations, 2022). The reference countries are chosen from Human Development Index (HDI) report (UNDP, 2022) and Body Mass Index (BMI) report (Abarca-Gómez et al., 2017). These datasets are all related to the topic which I would like to discuss.

There are four sections I would like to discuss in the visualisation. First of all, on graph 1-1 & 1-2, it can be seen that people in North America, Europe and Australia were taller than other areas. The reason why I chose 2014 is because it was the latest year from the dataset. Therefore, for other charts, I also chose 2014 as a reference year. Secondly, as to graph 2, the trend shows that the height of people in high HDI countries gradually reach a plateau. This could be a result from upper biological limits while human height does not continuously improve (Marck et al., 2017). Thirdly, graph 3 shows that the top 10 highest protein consumption countries mostly have animal protein which is consumed around 75 grams per day per person. When looking at graph 4, you can see that there is a positive correlation between meat consumption and average male height. However, males in Spain and Israel

could be outliers. Finally, graph 5 can be seen that in a high percentage of overweight people's countries, their average male BMI are mostly over 25 which is in the obesity range.

Theoretical frameworks

Ask a question:

I want to find out whether protein consumption is one of the factors for human height in order for the researchers to predict the future generation's development.

Search for information:

The data is from Our World in Data which collected wide aspects of information from technology to society. They are secondary and open and publicly accessible data where the website shows the original source. It is the high volume data because this involves many countries. It is also historical and latest because the data is updated year by year and released in recent years. Although the datasets are all independent, they can be grouped together because they are all related and can be visualised.

Structure the data:

- Graph 1-1 & 1-2
The original dataset is recorded from people who gave birth from 1896 to 1996. However, I think it is enough to show the latest mean height of people in the world. Therefore, I filtered the Year in 1996 and the Code without empty, and saved as a new sheet in order to read csv from R easily.
- Graph 2
Because it is hard to list all countries to see the trend of their height, I chose the 5 highest HDI countries (Germany, Iceland, Norway, Sweden and Switzerland) which represent adequate lifestyles as reference. The original data is only two columns, but in order to evaluate the trend, I divided the data by countries.
- Graph 3
I revised the original data in R. I removed empty cells, chose the year in 2014, descended the order, and finally chose only top 10 countries because it is not useful to see all the countries' information. Then I used matrix inverse in order to make a stacked plot, and cleared the data to be specific.
- Graph 4
This time I revised meat consumption data by choosing the top 10 countries where they consumed the most meat in 2014. Then I used `left_join` to combine `meat_consumption` and `men_height` by Entity. In order to make the data more meaningful, I only selected important columns, and changed the column names such as 'Entity' to 'Country'.
- Graph 5
I joined two datasets, one is from `male_BMI`, and the other is from `percentage_overweight`. First of all, I chose only the year in 2014 and without an empty Code. Then again, in order to make the graph more clear, I only chose the countries that consumed the most meat in the world to see whether they also have higher BMI and the percentage of overweight people.

Envision the answer (Grammar of Graphics) & Represent the visualisation

All graphs are transformed and become geometric objects. The charts are designed from Cartesian coordinates. Legend, title and caption are the must. Below is how I design the

charts. Please refer to page 2 to 4 to see the all visualisation, and refer to knowledge building for the outcomes.

- Graph 1-1 & 1-2: Map
 1. Use ggplot to paint a white map.
 2. Fill the map by Mean_female_height.
 3. Use scale_fill_viridis_c() for continuous data and suitable for colour blinder.
 4. Use theme_void() to clear any background colours.
- Graph 2: Line charts
 1. Use Year as x-axis, and add 5 lines as different countries.
 2. Scale the y-axis range in order to amplify and clear the trend.
 3. The colours are designed from ColorBrewer, and all colours are easy to read.
 4. Use grid.arrange to combine the two line charts.
- Graph 3: Stacked plot
 1. Use barplot to visualise the stacked plot.
 2. Scale y-axis in order to put the legend without covering the bar plot.
 3. Choose blue and red as bar colours for colour blind friendly.
- Graph 4: Scatter plot
 1. Use colours by country, but set the colour palette as 'Paired'.
 2. Set geom_point size as 6 in order to see the dots clearly.
 3. Use \n to not make the sentence too long.
- Graph 5: Tree map
 1. Fill the colours by Mean_male_BMI because the data is continuous.
 2. It is hard to distinguish the size of the areas, so I put the percentage data as a label and add it to the area.
 3. The words are white with italic font and placed in the centre.

Tell a story:

In 2014, people aged 18 in North America, Europe and Australia were taller than other countries. However, when we look deeper to see the 5 highest HDI countries which were all from Europe in 2014, it can be found that the height increased slowly and became a plateau. This can be seen as 'upper biological limits' (Marck et al., 2017). In the top 10 countries of people who consumed the most daily protein in 2014, mostly they ate animal protein, but plant protein is also consumed with certain percentages. Then, when we choose the top 10 countries where men ate the most meat and their average male height, there is a positive correlation. Finally, when we use the same countries to see the percentage of overweight people, they all had relatively high percentages. These graphs show that human height may be affected by animal protein but also cause obesity.

Accessibility

The meaning of accessibility in visualisation which I read from 'Data visualisation: A handbook for data driven design' is 'to remove design-related obstacles faced by your viewers when undertaking this process of understanding' (Kirk, 2016). I will cite all information from this book to answer 'what accessibility means in visualisation'.

The most important thing is to understand your audiences. For example, what they know, what they do not know, and what they need to know (Kirk, 2016). This requires visualisation designers to understand the audiences' capacity for learning new knowledge (Kirk, 2016). Nevertheless, a complex topic of visualisation could be oversimplified, or a visualisation with too much information can make it too complex and lead to unclear explanations (Kirk, 2016). Time spending is also an important issue for data visualisation. If the viewer cannot quickly perceive insight from the visualisation, they may have no patience to read the information in it and skip it. Therefore, it is necessary to understand whether your audience is willing to spend more time on evaluating the visualisation (Kirk, 2016). Moreover, we should consider people who have other sight situations such as colour blindness, visually impaired, and even blindness. For instance, when we use grammar of graphic, we try to avoid putting red and green together for colour blind people. The graph can also zoom in and out to make visually impaired people easily see some points clearly. There is little visualisation designed for blind people. However, tactile heatmaps (Engel et al., 2021) help blind people gain information more easily.

My visualisation is designed for people who would like to gain information about height, protein and obesity. They initially understand that eating protein can be a factor of height. However, they do not know they could have met the limited height because of upper biological limits (Marck et al., 2017). They also do not know the more meat they consume the more fat they intake. It is beyond the scope of the course for me to create tactile heatmaps (Engel et al., 2021) for blind people. It is also hard for me to create an interaction visualisation in a limited time. However, I use `scale_fill_viridis_c()` in the map, design colours from Coolers (<https://coolers.co/palettes/trending>) in line chart, set red and blue colours in the stacked plot, and use `scale_colour_brewer(palette='Paired')` in scatter plot that are set for colour blind people.

For colour design, I help colour blind people distinguish the categories. On the other hand, I set the limit of the y-axis from 160 to 180 (`scale_y_continuous(limits=c(160, 190))`) in order to make each line be seen clearly. Viewers can compare the line and the legend to match the country. As for the tree map, because the square size of each country is similar, it is hard to distinguish the percentage of overweight people in different countries. This may lead to viewers spending more time finding the important message in the tree map. Therefore, I create a new column for a label including country name and the percentage of overweight people. I also separate the name and percentage by '\n'. Then, I use subtitles on the right bottom side to explain what information is showing on the tree map. Finally, this helps viewers understand the tree map more.

Visualisation choice

According to the topic 'Whether protein consumption can influence human height?'. The goal is to make audiences understand the latest data which shows the trend of height and protein by visualisation. For extra information, I also create 2 plots in order to show the relationship between protein and fat.

There are 5 steps (plots) which explain the reason and the pros and cons.

Zoom	Out → In				
Graph	1-1 & 1-2	2	3	4	5
Plot type	World map	Line chart	Stacked plot	Scatter plot	Tree map
Data type	Categorical	Continuous	Categorical	Continuous & Continuous	Categorical
Reasons	To see the overall height in the world.	To see the highest HDI countries' trend of their height. The trend shows that the growth rate slows down.	To see and separate the rate of protein from meat and plants in top 10 protein consumption countries. Most countries intake animal protein more than plant-based protein.	To see the relationship between average male height and meat consumption which shows a positive association.	To see whether high meat consumption countries also have high BMI and high percentage of overweight people.
Positive aspects	It can be easily seen for high height countries by light colours.	The lines in the line chart can be seen clearly by different clear colours. The range of the y-axis is shortened in order to reduce the empty graph area.	The percentage of meat and plant-based protein are separated by red and blue which are contrasting colours and suitable for two categories.	Paired palette is used to separate 10 countries which are colour blind friendly. Because the dots do not influence each other. I do not have to use a translucent colour or position_jitter.	The tree map can use size and colour as two variables such as average male BMI and percentage of overweight people.
Negative aspects	Some countries' colours show grey because there is no data for them, so this is not a comprehensive map.	At some points, the lines are covered by another line so that it is hard to distinguish the difference. However, this chart is to see the overall trend, not for precise	There is no data shown on the bars to see the difference between each bar. However, this is an overall stacked plot to see people's main protein source is still from meat.	There are too many colours in this plot so it needs some time to match the colour and the country. However, this just shows the trend that the higher average male height of the	The size of areas are too similar to distinguish. However, I put the percentage text on the area.

		analysis.		country intake more meat.	
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There are two possible alternatives for stacked bar plot and tree map. Stacked plot can be changed to clustered bar chart, the left bar chart is red for meat protein and the right bar plot is for plant protein. This kind of graph can also display the difference between meat and plant protein easily because there are only two variables. On the other hand, the tree map can be altered to bar chart and line chart. For example, the y-axis of the bar chart shows the percentage of overweight people, and the x-axis shows different countries. Then, there is a line plotted on the chart showing the countries of average male BMI, and the label of BMI is put to the right side of the grid.

Implications and Improvements

Visualisations are not neutral because the data is collected from humans who have naturally unconscious bias. Therefore, when people represent the visualisations, the displayed value is wrong. It is inevitable to avoid. However, we can collect data from websites we trust such as the government or large nonprofit organisations to collect high volume and accurate data.

There are two examples I would like to talk about for inaccurate data collection. One is self-report data. There is evidence showing that height misreporting can lead to affect classification into WHO BMI categories (Mai et al., 2017). Height misreporting is usually found in women, some people's height are underestimated, and some people's height are overestimated (Mai et al., 2017). Researchers usually use self-reported height to investigate population obesity research (Mai et al., 2017). Hence, this self-reported data can lead to inaccurate conclusions which may influence future policies. Another example is the data which is collected by other people. For sedated patients, their height and weight are recorded by critical care staff. The evidence shows that even if the measurement is obtained by others not by themselves, the values could be wrong. Dunne and Hickey (2012) mentioned that the inaccurate height and weight collected by critical care staff can lead to wrong drug dose, inappropriate ventilatory settings, underfeeding or overfeeding. Patients need to have accurate medical treatment. The inaccurate record could lead to poor body condition or disease worsening.

There are some grammar graphics that could misinform the audience. As for graph 1-1 and 1-2, if I used non-uniform colour maps, the colour blinders might not distinguish the colours in each world area. For graph 2, I truncate axes so that the comparisons can be seen much starker. People usually read line charts from absolute zero, but the line charts I design are not for checking each average height every year. Therefore, I think truncating axes is reasonable to let audiences see the lines clearly and also see the overall trend. For scatter plots such as graph 4, some people use dot size to compare amounts, but actually it is hard to compare how big the dots are. For example, the dot A size might be 2.5 cm in diameter, and the dot B size might be 1 cm in diameter. However, we cannot see at a glance that the dot A size is 2.5 bigger than the dot B.

If there is any visualisation I would like to improve, I would choose graph 2, 3, 4, 5. As for graph 2, the y-axis range could be longer from 130 to 190. Although the average height in the line charts from 1975 to 2000 looks like it has decreased, the data shows the difference of height in 25 years is within 1 centimetre. For graph 3, I would add more variables in order to use ggplot to plot stacked plots because ggplot can do more customise such as putting values on the chart without writing too much code. For graph 4, I would select more countries in order to make the plot more persuasive. In addition, I will remove the legend and union the colour to black with translucency so that if the dots are at the same value, they will not make audiences confused or feel noise. This improvement is not to explain different countries with different values, but to understand if there is a trend for meat consumption and height or not. Finally, for graph 5, I will transfer the tree map to the bar chart with a line chart. The bar chart is for the percentage of overweight people, and the line chart is for the average male BMI. In addition, the chart can also add an overall male average BMI in the world to see the gap between countries' average male BMI and the overall world male BMI.

Reference

1. Dataset reference

- Rodriguez-Martinez, A., Zhou, B., Sophiea, M. K., Bentham, J., Paciorek, C. J., Iurilli, M. L., Carrillo-Larco, R. M., Bennett, J. E., Di Cesare, M., Taddei, C., Bixby, H., Stevens, G. A., Riley, L. M., Cowan, M. J., Savin, S., Danaei, G., Chirita-Emandi, A., Kengne, A. P., Khang, Y.-H., ... Ezzati, M.. (2020). Height and body-mass index trajectories of school-aged children and adolescents from 1985 to 2019 in 200 countries and territories: a pooled analysis of 2181 population-based studies with 65 million participants. *The Lancet*, 396(10261), 1511–1524. [https://doi.org/10.1016/s0140-6736\(20\)31859-6](https://doi.org/10.1016/s0140-6736(20)31859-6)
- UNDP. (2022). Human Development Report 2021-22. *UNDP (United Nations Development Programme)*. <http://report.hdr.undp.org>
- Abarca-Gómez, L., Abdeen, Z. A., Hamid, Z. A., Abu-Rmeileh, N. M., Acosta-Cazares, B., Acuin, C., ... & Cho, Y. (2017). Worldwide trends in body-mass index, underweight, overweight, and obesity from 1975 to 2016: a pooled analysis of 2416 population-based measurement studies in 128· 9 million children, adolescents, and adults. *The lancet*, 390(10113), 2627-2642.
- Food and Agriculture Organization of the United Nations. (2022, December 23). Food Balances (2010-). *Food and Agriculture Organization of the United Nations*. <https://www.fao.org/faostat/en/#data/FBS>

2. Literature review reference

- Mosites, E., Aol, G., Otiang, E., Bigogo, G., Munyua, P., Montgomery, J. M., ... & Thumbi, S. M. (2017). Child height gain is associated with consumption of animal-source foods in livestock-owning households in Western Kenya. *Public health nutrition*, 20(2), 336-345.
- de Lamas, C., de Castro, M. J., Gil-Campos, M., Gil, Á., Couce, M. L., & Leis, R. (2019). Effects of dairy product consumption on height and bone mineral content in children: a systematic review of controlled trials. *Advances in Nutrition*, 10(suppl_2), S88-S96.
- Ismail, B. P., Senaratne-Lenagala, L., Stube, A., & Brackenridge, A. (2020). Protein demand: Review of plant and animal proteins used in alternative protein product development and production. *Animal Frontiers*, 10(4), 53-63.
- Lin, Y., Bolca, S., Vandevijvere, S., De Vriese, S., Mouratidou, T., De Neve, M., . . . Huybrechts, I. (2011). Plant and animal protein intake and its association with overweight and obesity among the Belgian population. *British Journal of Nutrition*, 105(7), 1106-1116. doi:10.1017/S0007114510004642
- Morency, M. E., Birken, C. S., Lebovic, G., Chen, Y., L'Abbé, M., Lee, G. J., ... & TARGet Kids! Collaboration. (2017). Association between noncow milk beverage consumption and childhood height. *The American Journal of Clinical Nutrition*, 106(2), 597-602.
- Berrazaga, I., Micard, V., Gueugneau, M., & Walrand, S. (2019). The role of the anabolic properties of plant-versus animal-based protein sources in supporting muscle mass maintenance: a critical review. *Nutrients*, 11(8), 1825.

- Segovia-Siapco, G., Khayef, G., Pribis, P., Oda, K., Haddad, E., & Sabaté, J. (2019). Animal Protein Intake Is Associated with General Adiposity in Adolescents: The Teen Food and Development Study. *Nutrients*, 12(1), 110. <https://doi.org/10.3390/nu12010110>
- Bujnowski, D., Xun, P., Daviglus, M. L., Van Horn, L., He, K., & Stamler, J. (2011). Longitudinal association between animal and vegetable protein intake and obesity among men in the United States: the Chicago Western Electric Study. *Journal of the American dietetic association*, 111(8), 1150-1155.
- Halkjær, J., Olsen, A., Overvad, K., Jakobsen, M. U., Boeing, H., Buijsse, B., ... & Tjønneland, A. (2011). Intake of total, animal and plant protein and subsequent changes in weight or waist circumference in European men and women: the Diogenes project. *International journal of obesity*, 35(8), 1104-1113.
- Marck, A., Antero, J., Berthelot, G., Saulière, G., Jancovici, J. M., Masson-Delmotte, V., ... & Toussaint, J. F. (2017). Are we reaching the limits of homo sapiens?. *Frontiers in physiology*, 812.
- Kirk, A. (2016). *Data visualisation: A handbook for data driven design*. Sage.
- Engel, C., Müller, E. F., & Weber, G. (2021, June). Tactile Heatmaps: A Novel Visualisation Technique for Data Analysis with Tactile Charts. In *The 14th Pervasive Technologies Related to Assistive Environments Conference* (pp. 16-25).
- Mai, X., Sperrazza, J. N., Marshall, B. A., Hovey, K. M., & Wactawski-Wende, J. (2017). Inaccurate self-report of height and its impact on misclassification of body mass index in postmenopausal women. *Menopause (New York, NY)*, 24(5), 484.
- Dunne, K., & Hickey, S. (2012). Accuracy of height and weight estimation by critical care staff. *Critical Care*, 16(1), 1-189.

Appendix (R Code)

```
library(tidyverse)
library(ggplot2)
library(gridExtra)
library(treemapify)
library(dplyr)
```

#Height of women aged 18 around the world in 2014

```
women_height <- read.csv("women_height_revise.csv", header=TRUE)

world_map <- map_data("world")

ggplot(world_map, aes(x=long, y=lat, group=group))+
  geom_polygon(fill="white",colour="white")+
  theme(panel.background=element_blank())+
  labs(title="World map", caption="maps package, R")

women_heightMap <- left_join(world_map,women_height,by=c("region"="Entity"))

women_heightMap_graph <- ggplot(women_heightMap, aes(long, lat, group=group))+
  geom_polygon(aes(fill=Mean_female_height), colour="white")+
  scale_fill_viridis_c()+
  theme_void()+
  labs(fill="Mean Height(cm)",
        title="Mean height of women aged 18 in 2014",
        caption="NCD RisC, Human Height (2017)")

women_heightMap_graph
```

#Height of men aged 18 around the world in 2014

```
men_height <- read.csv("men_height_revise.csv", header=TRUE)

ggplot(world_map, aes(x=long, y=lat, group=group))+
  geom_polygon(fill="white",colour="white")+
  theme(panel.background = element_blank())+
  labs(title="World map", caption="maps package, R")

men_heightMap <- left_join(world_map,men_height,by=c("region"="Entity"))

men_heightMap_graph <- ggplot(men_heightMap, aes(long,lat,group=group))+
  geom_polygon(aes(fill=Mean_male_height), colour="white")+
  scale_fill_viridis_c()+
  theme_void()+
  labs(fill="Mean Height(cm)",
```

```
title="Mean height of men aged 18 in 2014",  
caption="NCD RisC, Human Height (2017)"
```

```
men_heightMap_graph
```

#Annual change in average male height in 5 highest HDI countries

```
annual_change_men_height <- read.csv("men_height_HDI.csv", header=TRUE)  
  
annual_change_men_height_graph <- ggplot(annual_change_men_height, aes(x=Year))+  
  geom_line(aes(y=Norway, col="Norway"))+  
  geom_line(aes(y=Switzerland, col="Switzerland"))+  
  geom_line(aes(y=Iceland, col="Iceland"))+  
  geom_line(aes(y=Germany, col="Germany"))+  
  geom_line(aes(y=Sweden, col="Sweden"))+  
  labs(x="Year", y="Mean Height(cm)",  
    title="Annual change in average height of 18y males by year of birth in 5 highest HDI  
countries (1896~1996)")+  
  scale_y_continuous(limits=c(160, 190))+  
  scale_colour_manual(name="Legend",  
    values=c("Norway"="#8338ec",  
      "Switzerland"="#219ebc",  
      "Iceland"="#023047",  
      "Germany"="#ffb703",  
      "Sweden"="#fb8500"))
```

#Annual change in average female height in 5 highest HDI countries

```
annual_change_women_height <- read.csv("women_height_HDI.csv", header=TRUE)  
  
annual_change_women_height_graph <- ggplot(annual_change_women_height,  
aes(x=Year))+  
  geom_line(aes(y=Norway, col="Norway"))+  
  geom_line(aes(y=Switzerland, col="Switzerland"))+  
  geom_line(aes(y=Iceland, col="Iceland"))+  
  geom_line(aes(y=Germany, col="Germany"))+  
  geom_line(aes(y=Sweden, col="Sweden"))+  
  labs(x="Year", y="Mean Height(cm)",  
    title="Annual change in average height of 18y females by year of birth in 5 highest HDI  
countries (1896~1996)",  
    caption="NCD RisC (2017)")+  
  scale_y_continuous(limits=c(150, 180))+  
  scale_colour_manual(name="Legend",  
    values=c("Norway"="#8338ec",  
      "Switzerland"="#219ebc",  
      "Iceland"="#023047",  
      "Germany"="#ffb703",  
      "Sweden"="#fb8500"))
```



```
grid.arrange(annual_change_men_height_graph, annual_change_women_height_graph,  
ncol=1)
```

#animal and plant-based protein

```
animal_plant_protein <-  
read.csv("daily-protein-supply-from-animal-and-plant-based-foods.csv", header=TRUE)
```

```
animal_plant_protein[animal_plant_protein == " "] <- NA
```

```
animal_plant_protein <- animal_plant_protein[!is.na(animal_plant_protein$Code) &  
      animal_plant_protein$Year==2014, ]
```

```
animal_plant_protein <- animal_plant_protein[order(animal_plant_protein$Animal_products,  
decreasing=TRUE),]
```

```
animal_plant_protein <- animal_plant_protein[1:10,]
```

```
animal_plant_protein <- t(animal_plant_protein) #matrix inverse
```

```
colnames(animal_plant_protein) <- animal_plant_protein[1, ] #change column name as  
country
```

```
animal_plant_protein <- animal_plant_protein[4:5,] #choose only protein information
```

```
barplot(animal_plant_protein,  
  main = "Daily protein supply from animal and plant-based foods in 2014",  
  sub = "Source from Food and Agriculture Organization of the United Nations (2022)",  
  xlab = "Country",  
  ylab = "Grams per day per capita",  
  ylim = c(0,200),  
  col = c("Red","Blue"))
```

```
legend("topright",  
  c("Plant protein","Animal protein"),  
  fill = c("Blue","Red"))
```

#Animal protein consumption versus average male height

```
meat_consumption <- read.csv("meat_consumption.csv", header=TRUE)
```

```
male_height_protein <- left_join(meat_consumption, men_height, by=c("Entity"="Entity"))
```

```
male_height_protein <- male_height_protein %>% select(1, 3, 4, 7)
```

```
colnames(male_height_protein)[1] <- "Country"
```

```
colnames(male_height_protein)[2] <- "Year"
```

```
View(male_height_protein)
```

```
male_height_protein_graph <- ggplot(male_height_protein, aes(Mean_male_height, kg.pp,
col=Country))+
  geom_point(size=6)+
  scale_colour_brewer(palette='Paired')+
  labs(x='Average male height(cm)', y='Meat consumption (kilograms per capita per year)',
  title="Average male height by top 10 countries where men ate the most meat in 2014",
  caption="NCD RisC (2017)\nFood and Agriculture Organization of the United
Nations(2022)")
```

```
male_height_protein_graph
```

[#Choose top 10 meat consumption of countries to see the relation between their mean BMI and the percentage of overweight people](#)

```
male_BMI <- read.csv("male_BMI.csv", header=TRUE)
```

```
male_BMI[male_BMI == " "] <- NA
```

```
male_BMI <- male_BMI[!is.na(male_BMI$Code) &
male_BMI$Year==2014, ]
```

```
male_BMI <- male_BMI[male_BMI$Entity %in% c("Australia", "United States", "Argentina",
"Samoa", "Brazil", "Israel", "Bahamas", "Spain", "Saint Lucia", "Saint Vincent and the
Grenadines"),]
```

```
percentage_overweight <- read.csv("percentage_overweight.csv", header=TRUE)
```

```
percentage_overweight[percentage_overweight == " "] <- NA
```

```
percentage_overweight <- percentage_overweight[!is.na(percentage_overweight$Code) &
percentage_overweight$Year=="2014", ]
```

```
percentage_overweight <- percentage_overweight %>% select(1,4)
```

```
overweight_BMI <- left_join(male_BMI, percentage_overweight,by=c("Entity"="Entity"))
```

```
overweight_BMI <- overweight_BMI[!is.na(overweight_BMI$Percentage_of_overweight),]
```

```
overweight_BMI$label <- paste(overweight_BMI$Entity,
overweight_BMI$Percentage_of_overweight, sep = "\n")
```

```
ggplot(overweight_BMI, aes(area=Percentage_of_overweight, fill=Mean_male_BMI,
label=label))+
```

```
geom_treemap()+  
geom_treemap_text(fontface="italic", colour="white", place="centre")+  
labs(title="Average male BMI by percentage of overweight in countries of high meat  
consumption by males in 2014", fill="Average male BMI",  
caption="The area shows country name and percentage of overweight people")
```